

Di Hou

National University of Singapore

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Appointments

Research Fellow | Advisor: Prof. Kim-Chuan Toh 2026 - Present
Department of Mathematics, National University of Singapore, Singapore

Education

Ph.D. in Mathematics | Advisor: Prof. Kim-Chuan Toh 2022 - 2026
Department of Mathematics, National University of Singapore, Singapore

Bachelor of Science in Mathematics (Honors) 2018 - 2022
Department of Mathematics, Jilin University, Changchun, China

Research Interests

- Optimization
- Scientific computing

Publications and Preprints

* Corresponding author.

- [1] Nachuan Xiao, **Di Hou**^{*}, Kim-Chuan Toh. Dual Parameterization for Nonconvex Constrained Optimization. **SIAM Journal on Optimization**, submitted.
- [2] **Di Hou**, Thai P.D. Nguyen, Kim-Chuan Toh, Guanyi Wang. Sparsity-Cone SDP Relaxations and Applications to Variable Fixing for Sparse Quadratic Programs. **Operations Research**, submitted.
- [3] **Di Hou**, Tianyun Tang, Kim-Chuan Toh, Shiwei Wang. On the efficient computation of proximal operators of affine-constrained nonconvex functions. **Mathematics of Operations Research**, major revision.
- [4] **Di Hou**, Tianyun Tang, Kim-Chuan Toh. A low-rank augmented Lagrangian method for Polyhedral-SDP and Moment-SOS relaxations of polynomial optimization. **Mathematical Programming** (2026), pp. 1–53.
- [5] **Di Hou**, Tianyun Tang, Kim-Chuan Toh. RiNNAL+: a Riemannian ALM solver for SDP-RLT relaxations of mixed-binary quadratic programs. **Mathematical Programming Computation** (2026), pp. 1–57.
- [6] **Di Hou**, Tianyun Tang, Kim-Chuan Toh. A Low-rank ALM for doubly nonnegative relaxations of mixed-binary quadratic programs. **Operations Research** 74(3) (2025), pp. 1626–1647.
- [7] **Di Hou**, Ling Liang, Kim-Chuan Toh. A sparse smoothing Newton method for solving discrete optimal transport problems. **ACM Transactions on Mathematical Software**, 50 (2024), pp. 1-26.

- [8] Jinda Yang, Haiming Song, Xinxin Li, **Di Hou**. Block mirror stochastic gradient method for stochastic optimization.
Journal of Scientific Computing, 94 (2023).

Teaching

- MA 4260 Stochastic Operations Research, semester 2, AY 2025/26
- DSA2102 Essential Data Analytics Tools: Numerical Computation, semester 2, AY 2024/25
- MA 3236 Non-Linear Programming, semester 1, AY 2024/25
- DSA3102 Essential Data Analytics Tools: Convex Optimisation, semester 2, AY 2023/2024
- DSA3102 Essential Data Analytics Tools: Convex Optimisation, semester 1, AY 2023/2024

Talks

- *RiNNAL-X: Low-Rank ALM for Structured SDP Relaxations*, Computational Robotics Group Seminar (online) Harvard University, Cambridge, MA, U.S., June 2026.
- *RiNNAL-POP: A Low-rank Augmented Lagrangian Method for Polyhedral-SDP and Moment-SOS Relaxations of Polynomial Optimization*, SIAM Conference on Optimization (OP26), Edinburgh, United Kingdom, June 2026.
- *RiNNAL-POP: A Low-rank Augmented Lagrangian Method for Polyhedral-SDP and Moment-SOS Relaxations of Polynomial Optimization*, Mini Workshop on Optimization, National University of Singapore, Dec. 2025.
- *RiNNAL+: a Riemannian ALM solver for SDP-RLT relaxations of MBQP*, The 8th PKU Workshop for Outstanding PhD Students in Computational and Applied Mathematics, Peking University, Beijing, China, Nov. 2025.
- *RiNNAL+: a Riemannian ALM solver for SDP-RLT relaxations of MBQP*, International Conference on Continuous Optimization (ICCOPT) 2025, Los Angeles, California, U.S., July 2025.
- *RiNNAL: A Low-Rank ALM for Doubly Nonnegative Relaxations of Mixed-Binary QP*, The 9th NUS Graduate Symposium in Mathematics, National University of Singapore, May 2025.
- *A Sparse Smoothing Newton Method for Solving Discrete Optimal Transport Problem*, NUS-SJTU PhD Forum, National University of Singapore, Nov. 2024.

Recognitions

- Top Graduate Researcher Award, Faculty of Science, NUS, 2026.
- Best Graduate Researcher Award, Department of Mathematics, NUS, 2026.
- PhD Conference Award, Faculty of Science, NUS, 2025.
- President's Graduate Fellowship, NUS, Singapore, 2022-2026.
- Graduate Tutor Merit Award, NUS, Singapore, 2024 (Sem. 1 and Sem. 2), 2025 (Sem. 1).
- Top Ten Undergraduate Student Award, Changchun Province, China, 2022.
- Top Ten Undergraduate Student Award, Jilin University, China, 2022.

- National First Prize Award, Chinese Mathematics Competitions, 2021.
- National Scholarship, China, 2018-2022.

Service

Referee for Journals

- Optimization Methods and Software
- ACM Transactions on Mathematical Software
- Optimization and Engineering
- Numerical Algebra, Control and Optimization

University services

- President of NUS SIAM Student Chapter, Jul. 2025 – Present.
- Secretary of NUS SIAM Student Chapter, Jul. 2024 – Jul. 2025.
- Vice president of NUS Mathematics Graduate Society, Jul. 2025 – Present.

Selected events

- Organizer, NUS-SDU-TJU joint math PhD forum, May 2026.
- Organizer, 10th NUS Graduate Symposium in Mathematics, Apr. 2026.
- Organizer, 15th Symposium of the SIAM Student Chapter at NUS, Mar. 2026.
- Chair, NUS 120th Anniversary Math & Statistics Forum and Alumni Homecoming, Oct. 2025.
- Organizer, 14th Symposium of the SIAM Student Chapter at NUS, Mar. 2024.